

LABORATORY PROFILE

Physics-powered precision healthcare

Advanced solutions spanning radiation transport, dosimetry, medical imaging, artificial intelligence, and scientific validation.

RADIATION TRANSPORT

DOSIMETRY

MEDICAL IMAGING

AI

VALIDATION

TWO CO-DEVELOPMENT OFFERS

01 PRECISION DOSIMETRY

Monte Carlo simulation for accurate, patient-specific radiation dose calculation.

02 CLINICAL SECOND OPINION

Medical imaging AI designed to support physicians, not replace clinical judgement.

PARTNERSHIP MODEL

Needs & Data → Proof of Concept → Technical Validation →
Clinical Validation → Integration & Scale-Up



**Precision healthcare solutions powered by physics,
medical imaging, AI, and systems engineering**

OFFER 01

Monte Carlo simulation for precision dosimetry

Objective: improve the precision and accuracy of patient-specific dose calculations in ionising radiation therapy.

PROPOSED SCOPE

SOURCE

Radiation-source and LINAC-head modelling

PATIENT

CT-based patient geometry modelling

DOSE

3D dose distribution and DVH calculation

BENCHMARK

Measurement and treatment-planning comparison

QA

Independent secondary dose calculation

DEPLOY

Computational optimisation for clinical implementation



RESEARCH FOUNDATION

Photon-beam output factors for small fields
Virtual photon energy fluence modelling
Monte Carlo dose-calculation engines

EXPECTED OUTPUT

**Validated Monte Carlo simulation engine +
quality-assurance protocol**

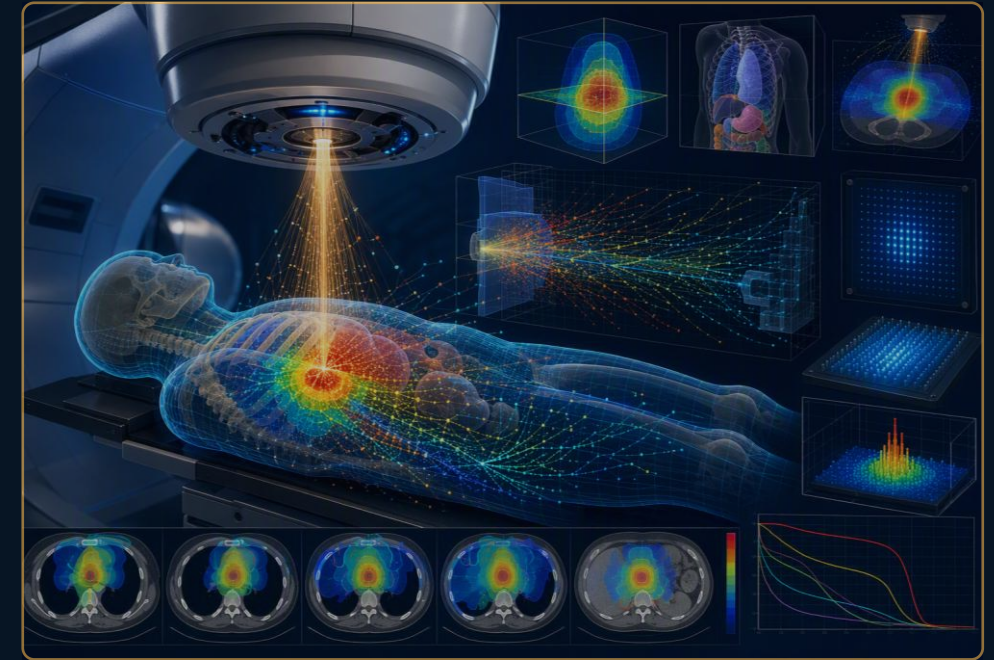
OFFER 02

Medical imaging AI for clinical second opinion

Clinical decision-support tools that assist physicians while preserving clinical judgement and accountability.

END-TO-END DEVELOPMENT SCOPE

- | | |
|---|---|
| 01 Curate, anonymise & annotate data | 02 Segment, classify & predict risk |
| 03 Explain predictions & visualise uncertainty | 04 Validate externally & assess bias |
| 05 Build dashboards & integrate workflows | 06 Assess patient-specific CT dose |



RESEARCH TRACK RECORD

99.59%

reported test accuracy
CNN urinary-stone
classification

CT DOSE

Automated water-equivalent
diameter for patient-specific
estimation

EXPECTED OUTPUT

**AI prototype + decision-support dashboard +
validation report**

LAB ROLE Physics modelling • AI methodology • Dosimetric & clinical validation • Scientific publications • Talent development